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PAPER

Physical activity and its relationship with obesity, hypertension and diabetes in urban and rural Cameroon

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OBJECTIVE: To evaluate and compare physical activity patterns of urban and rural dwellers in Cameroon, and study their relationship with obesity, diabetes and hypertension.
METHODS: We studied 2465 subjects aged ≥ 15 y, recruited on the basis of a random sampling of households, of whom 1183 were urban dwellers from Yaoundé, the capital city of Cameroon and 1282 rural subjects from Bafut, a village of western Cameroon. They all had an interviewer-administered questionnaire for the assessment of their physical activity and anthropometric measurements, blood pressure and fasting blood glucose determination. The procedure was satisfactorily completed in 2325 (94.3%) subjects. Prevalences were age-adjusted and subjects compared according to their region, sex and age group.
RESULTS: Obesity was diagnosed in 17.1 and 3.0% urban and rural women, respectively ($P<0.001$), and in 5.4 *vs* 1.2% urban and rural men, respectively ($P<0.001$). The prevalence of hypertension was significantly higher in urban *vs* rural dwellers (11.4 *vs* 6.6% and 17.6 *vs* 9.1% in women and men, respectively; $P<0.001$). Diabetes was more prevalent in urban compared to rural women ($P<0.05$), but not men. Urban subjects were characterized by lower physical activity ($P<0.001$), light occupation, high prevalence of multiple occupations, and reduced walking and cycling time compared to rural subjects. Univariate analysis showed significant associations between both physical inactivity and obesity and high blood pressure. The relationship of physical inactivity with hypertension and obesity were independent in both urban and rural men, but not in women. Body mass index, blood pressure and glycaemia were higher in the first compared with the fourth quartiles of energy expenditure.
CONCLUSION: Obesity, diabetes and hypertension prevalence is higher in urban compared to rural dwellers in the populations studied. Physical activity is significantly lower and differs in pattern in urban subjects compared to rural. Physical inactivity is associated with these diseases, although not always significant in women.
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Introduction

Cross-sectional epidemiological studies and interventions have demonstrated the benefit of physical activity in the primary prevention of diabetes mellitus, coronary heart disease and hypertension, although evidence is sparse in women.[1–3] The majority show an association between inactivity and these non-communicable diseases (NCDs); however, it remained unclear whether high levels of physical activity had greater protective effect than moderate physical activity.

Sub-Saharan Africa (SSA) is known to be entering the epidemiological transition[4] with the burden of cardiovascular diseases and type 2 diabetes mellitus increasing as the population is ageing.[5] There is a reduction of mortality, particularly childhood mortality, from communicable diseases and undernutrition, and decreasing fertility, in a context of rapid urbanization, overnutrition and adoption of

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Westernized lifestyle that characterizes the ongoing nutritional transition.[6,7] Urbanization appears to be associated with extreme changes in dietary habits, psychological stress, subsistence means and physical activity.[8] Evidence from the literature suggests that, in general, physical activity levels are much higher in Africa than in the Western world, especially in populations living in rural areas.[9] In energy balance studies where the doubly labelled water technique has been used, energy expenditure related to physical activity was higher in rural Gambians than Western populations.[10,11] Studies in children and adolescents also provide some support to the general observation that physical activity levels decrease with urbanization in African populations.[12]

Likewise, great urban–rural differences in the prevalence and incidence of various cardiovascular disorders and diabetes mellitus have been reported in Africa, but whether this gradient is attributable to physical activity has rarely been shown.[13,14] One reason suggested for the lack of a clear association is the absence of precision in the measurement of physical activity to define risk.[15] Where objective and accurate measures of physical activity were performed in sub-Saharan Africa, the target sample size was limited, the main focus was undernutrition or energy balance, and physical activity levels were seldom analyzed in relation with NCDs.[16,17]

We therefore used a piloted and validated culture-sensitive questionnaire[18] to study the physical activity patterns of two urban and rural African communities. We studied the relationship between current level of physical activity and obesity, hypertension and diabetes with a view to developing an intervention programme.

Methods

We conducted a cross-sectional population-based survey to determine the prevalence of obesity, hypertension and diabetes and their relationship to physical inactivity in two settings: an urban area, Biyem-Assi, situated in the capital city of Cameroon, made up mainly of civil servants, businessmen and students; and a rural setting, the Bafut health district in the fertile highlands of north-western Cameroon, where subsistence farming is the main source of income. These settings are the sites of the Cameroon Essential Non-communicable disease Health Intervention Project (ENHIP),[19] a project that was designed to pilot and implement guidelines for the treatment of epilepsy, asthma, diabetes and hypertension at primary care level. The choice of these sites allowed easy referral of newly diagnosed cases for appropriate care.

Study population

The study population consisted of 1183 urban dwellers from Biyem-Assi (Yaoundé) and 1282 rural dwellers from Bafut aged ≥ 15 y from households of 1–13 people. They were recruited on the basis of a random sampling of households following a preliminary census of households in the study area. In an effort to include all eligible household residents, three separate visits were made to each household. The participation rate was 96 and 98% in the urban and the rural populations, respectively.

Procedure

Home visits were organized for the collection of data by trained survey workers.

Questionnaire. A questionnaire comprising sections on personal information, self report of diabetes, hypertension or ischaemic heart disease, and alcohol and tobacco consumption was used. The Sub-Saharan Africa Activity Questionnaire (SSAAQ) was administered to all the participants to assess their habitual physical activity. This questionnaire includes the assessment of the past year's occupation, walking and leisure-time activities and was pre-tested and validated against heart rate monitoring and accelerometer in the target population.[18] This questionnaire has been shown to be highly reproducible with an inter-interview rank correlation of $\rho = 0.95$ ($P < 0.001$), and a repeatability coefficient of 0.46–1.46. Total energy expenditure from the questionnaire significantly correlated to heart rate monitoring ($\rho = 0.41$–0.63; $P < 0.05$) and accelerometer measures ($\rho = 0.60$–0.74; $P < 0.01$).[18]

Measurements. Height was measured to the nearest 0.5 cm, weight to the nearest 0.5 kg, waist and hip circumference to the nearest 0.5 cm, and BMI (weight/height^2 in kg/m^2) and waist/hip ratio calculated as previously described.[13] The average of two blood pressure measurements performed following the procedure recommended by the British Hypertension Society[20] on the right arm of the subjects in a sitting position, after a minimum 10 min rest using a standard mercury sphygmomanometer with appropriate sized cuff was used in the analyses. Blood glucose was measured between 7 and 9 am after a minimum 10 h overnight fast using a Hemocue blood glucose analyser (Angelholm, Sweden) on capillary blood samples.

Physical activity calculations and interpretation

Frequency and duration were computed for each reported activity, and the metabolic cost calculated using Ainsworth *et al*'s compendium.[21,22] This metabolic cost is expressed as a MET score, which is the ratio of the working metabolic rate of an activity divided by the resting metabolic rate. Energy expenditure related to physical activity was therefore computed by multiplying the MET score by the number of hours spent in each activity. Occupation was categorized into three groups — light, moderate and intense — according to the type of activity performed, with an estimated metabolic cost of 1.5, 2.5 and 6.0 METs respectively. Estimates of total energy expenditure were computed assuming that the time not reported was spent at the resting metabolic rate (1 kcal/kg/h). Results are expressed as hours of activity or METs expended for occupation, walking/cycling, leisure-time and total activity per week. We included in the analysis only subjects who reported ≤ 133 weekly hours of activity; reports of > 133 h/week (19 h/day) of activity were considered unrealistic and probably erroneous and therefore excluded from the analysis.

Definitions

Obesity. Subjects were classified as obese if they had a body mass index (BMI) $\geq 30\,kg/m^2$, and overweight for a BMI ≥ 25 and $< 30\,kg/m^2$.[23]

Hypertension. Hypertension was defined according to WHO 1993 recommendations based on current treatment by hypotensive medication or a systolic blood pressure (SBP) ≥ 160 mmHg or diastolic blood pressure (DBP) ≥ 95 mmHg. Borderline hypertension was defined as $140 < SBP < 160$ or $90 < DBP < 95$ mmHg.[24]

Diabetes. We used fasting capillary glucose ≥ 6.1 mmol/l (or hypoglycaemic treatment) to define diabetes, together with ≥ 5.6 and < 6.1 mmol/l to define impaired fasting glycaemia.[25]

Statistical analysis

Statistical analyses were performed using the Statistical Package for Social Sciences (SPSS) version 7.5 for windows and STATA version 5. Adjustments to estimates were made for the effect of clustering, as the sample units were households. Analyses were stratified by area and sex. Comparisons between urban and rural men and women were done by the median test for continuous variables, and chi-square test for quantitative variables. Prevalence of overweight, obesity, impaired fasting glycaemia, diabetes, borderline hypertension and hypertension was standardized to the Cameroon urban and rural population projections based on the 1987 national demographic survey.[26] For the purpose of comparison, disease prevalence was age-adjusted to the Cameroon urban and rural population. Age standardization was by the direct method and confidence intervals calculations for population surveys were used. Associations between obesity, diabetes, and hypertension and physical activity were examined by univariate and multivariate regression analyses. The study population was further subdivided into quartiles of physical activity energy expenditure in METs, which were compared by means of Student's *t*-tests. A *P* value < 0.05 was considered statistically significant.

Ethics

The National Ethical committee of the Ministry of Public Health of Cameroon approved the study. All subjects gave verbal informed consent before they were enrolled in the study and the authors followed the principles of the Declaration of Helsinki on biomedical research involving human subjects.

Results

Characteristics of the population

A total of 1417 women and 1048 men distributed across the urban and rural settings participated in the study (Table 1).

Disease prevalence

Obesity. A BMI $\geq 25\,kg/m^2$ was found in less than 10 and 2% of rural women and men, respectively, whereas in the urban population, 14% of men and more than 30% of women were overweight or obese. Obesity was an exceptional finding in the rural population (0.5% in men and 3.0% in women; Table 2).

Table 1 Characteristics of the study population (unadjusted)

	Women			*Men*		
	Urban	*Rural*	P	*Urban*	*Rural*	P
n	658	759		525	523	
Age (y)[a]	27 (20–40)	46 (31–58)	<0.001	26 (20–42)	48 (34–60)	<0.001
Body mass index (kg/m^2)[a]	24.4 (21.8–27.7)	21.9 (20.0–24.0)	<0.001	22.9 (20.9–25.1)	21.0 (19.5–22.8)	<0.001
Waist-to-hip ratio[a]	0.80 (0.75–0.84)	0.84 (0.80–0.88)	<0.001	0.85 (0.81–0.89)	0.87 (0.83–0.90)	<0.001
Fasting glycaemia (mmol/l)[a]	4.8 (4.5–5.2)	4.5 (4.2–5.0)	<0.001	4.7 (4.4–5.1)	4.6 (4.1–5.1)	0.07 (NS)
Blood pressure (mmHg)[a]						
Systolic	119 (111–129)	116 (106–127)	<0.001	125 (117–137)	121 (111–134)	<0.001
Diastolic	76 (70–84)	75 (68–83)	0.02	81 (74–88)	79 (71–86)	0.22 (NS)
Current smoking (%)	4	14	0.03	14	32	0.004
Daily alcohol intake >30 g (%)	5	37	<0.001	17	59	<0.001
Daily energy expenditure:[b]						
in MET	36.1±0.3	45.6±0.4	<0.001	39.7±0.6	46.9±0.7	<0.001
in kJ	9990±122	10 588±122	<0.001	11 340±210	11 619±210	NS

[a]Mean (95% confidence interval); [b]mean±standard error.

Hypertension. High blood pressure was almost twice as prevalent in urban than rural subjects and with a significantly higher prevalence in men than women within the sites (Table 2).

Diabetes. The prevalence of diabetes mellitus and impaired fasting glycaemia was higher in urban *vs* rural women and men (Table 2).

Activity pattern

Valid reports of activity were recorded in 94.3% of the study population with a median total activity time reported of 55 h/week. Invalid reports were distributed as follows: 5.2 and 3.0% in urban and rural women, respectively, and 9.8 and 5.6% in urban and rural men, respectively.

Occupation. Among the population included in the analysis, 13.0% had no occupation and 13.7% reported more than one currently held occupation (multiple occupation). Multiple occupation was typically an urban finding. Intense occupational activity was reported in 2.7% of urban subjects compared with 79.6% of rural subjects ($P<0.001$), although urban subjects tended to spend more time at work in all the age groups (Tables 3 and 4).

Transportation. Bicycle was exceptionally reported as means of transportation to work in urban subjects. Mean time spent for walking at a brisk pace per day was 30–54 min

Table 2 Age-adjusted prevalence (95% confidence interval) of obesity, diabetes and hypertension in the study population to the Cameroon urban and rural population

	Women			*Men*		
	Urban	*Rural*	P	*Urban*	*Rural*	P
n	658	759		525	523	
Obesity						
Overweight	14.3 (11.3–17.3)	6.2 (4.0–8.5)	<0.001	8.6 (6.0–11.1)	1.1 (0.3–1.9)	<0.001
Obesity	17.1 (13.9–20.4)	3.0 (1.7–4.3)	<0.001	5.4 (3.2–7.5)	0.5 (0.1–0.9)	<0.001
Hypertension						
Borderline	8.6 (6.1–11.1)	5.2 (3.5–7.0)	0.004	14.2 (10.9–17.6)	7.3 (4.6–10.0)	<0.001
Hypertension	11.4 (8.9–14.0)	6.6 (4.8–8.3)	<0.001	17.6 (14.2–21.1)	9.1 (6.3–12.0)	<0.001
Diabetes mellitus						
Impaired fasting glycaemia	10.4 (7.8–13.0)	4.7 (2.8–6.6)	<0.001	4.4 (2.6–6.3)	4.9 (2.4–7.4)	0.671 (NS)
Diabetes	4.7 (2.6–6.8)	2.9 (1.5–4.4)	0.05	6.2 (3.7–8.9)	4.7 (2.5–6.9)	0.168 (NS)

Table 3 Activity pattern in women

	<30 y			*30–49 y*			*≥50 y*		
	Rural	*Urban*	P	*Rural*	*Urban*	P	*Rural*	*Urban*	P
Valid reports **(*n*)**	162	353		274	219		300	52	
Main occupation category (%)			<0.001			<0.001			<0.001
Light (A)	13.0	79.8		5.1	62.7		0.7	35.8	
Moderate (B)	6.5	3.8		4.0	15.5		1.3	15.1	
Intense (C)	63.0	0.0		88.7	1.7		84.7	9.4	
None	17.4	16.4		2.2	20.2		13.3	39.6	
Multiple occupation (%)	1.6	40.3	<0.001	0.7	12.0	<0.001	1.2	9.4	<0.001
Occupation (h/week)[a]	23.2±1.2	31.9±1.0	<0.001	23.3±0.8	28.0±1.5	0.004	16.0±0.6	19.9±3.0	0.02
Walking (min/day)[a]									
Slow pace	29±5	19±2	0.007	52±5	13±2	<0.001	64±4	19±5	<0.001
Brisk pace	45±4	8±1	<0.001	54±4	4±1	<0.001	30±3	2±1	<0.001
Cycling (min/day)[a]	9±6	0	—	6±4	0	—	2±2	0	—
Leisure-time activity (h/week)[a]									
Light	3.3±0.4	23.9±0.8	<0.001	2.6±0.3	23.9±1.1	<0.001	1.2±0.1	17.9±2.0	<0.001
Moderate	9.8±0.5	6.3±0.4	<0.001	11.7±0.5	7.8±0.7	<0.001	10.2±0.4	4.5±0.8	<0.001
Intense	3.0±0.4	2.6±0.3	0.81 (NS)	3.3±0.5	2.3±0.4	0.83 (NS)	1.7±0.3	3.6±1.3	0.001
Total	16.1±0.8	32.7±1.0	<0.001	17.7±0.7	34.0±1.2	<0.001	13.1±0.5	25.9±2.6	<0.001
Total daily expenditure[a]									
in MET	45.3±0.9	35.9±0.4	<0.001	48.9±0.8	36.6±0.5	<0.001	42.8±0.6	35.9±1.4	<0.001
in kJ	10955±262	9156±122	<0.001	11759±210	11043±210	0.02	9330±175	9959±437	0.17 (NS)

[a]Mean±standard error.

Table 4 Activity pattern in men

	< 30 y			30–49 y			≥ 50 y		
	Rural	*Urban*	P	*Rural*	*Urban*	P	*Rural*	*Urban*	P
Valid reports (***n***)	94	276		163	132		242	58	
Main occupation category (%)			< 0.001			< 0.001			< 0.001
Light (A)	14.0	75.2		12.7	61.5		5.8	49.2	
Moderate (B)	6.1	8.7		7.8	24.4		4.5	8.5	
Intense (C)	58.8	4.8		74.7	6.4		73.7	3.4	
None	21.1	11.3		4.8	7.7		16.0	39.0	
Multiple occupation (%)	0.0	43.2	—	2.4	10.9	0.004	1.2	3.4	0.50 (NS)
Occupation (h/week)[a]	30.0 ± 1.2	31.7 ± 1.1	0.42 (NS)	26.5 ± 1.5	40.2 ± 1.9	< 0.001	19.7 ± 1.1	22.8 ± 3.1	0.24 (NS)
Walking (min/day)[a]									
Slow pace	21 ± 6	17 ± 2	0.54 (NS)	31 ± 6	11 ± 3	0.003	43 ± 4	4 ± 2	< 0.001
Brisk pace	38 ± 4	17 ± 2	< 0.001	49 ± 6	12 ± 3	< 0.001	29 ± 3	2 ± 1	< 0.001
Cycling (min/day)[a]	5 ± 5	0	—	6 ± 4	2 ± 2	0.40 (NS)	3 ± 2	0	—
Leisure-time activity (h/week)[a]									
Light	2.2 ± 0.4	20.4 ± 1.0	< 0.001	3.7 ± 0.6	22.9 ± 1.4	< 0.001	2.4 ± 0.2	14.5 ± 2.0	< 0.001
Moderate	9.2 ± 0.9	5.0 ± 0.4	< 0.001	6.3 ± 0.5	4.2 ± 0.5	0.004	6.4 ± 0.5	4.2 ± 1.0	0.13 (NS)
Intense	4.3 ± 0.8	5.6 ± 0.5	0.06 (NS)	4.5 ± 0.7	3.4 ± 0.6	0.20 (NS)	3.4 ± 0.5	2.0 ± 0.7	0.42 (NS)
Total	15.6 ± 1.3	31.2 ± 1.2	< 0.001	14.5 ± 1.0	30.5 ± 1.6	< 0.001	12.2 ± 0.7	20.8 ± 2.5	< 0.001
Total daily expenditure[a]									
in MET	50.4 ± 1.8	40.6 ± 0.8	< 0.001	49.1 ± 1.4	40.6 ± 1.3	< 0.001	44.0 ± 1.0	33.3 ± 1.6	< 0.001
in kJ	12 510 ± 524	10 885 ± 227	0.002	12 283 ± 367	12 772 ± 419	0.36 (NS)	10 815 ± 262	10 134 ± 594	0.27 (NS)

[a]Mean ± standard error

in rural women compared with 2–8 min in urban women ($P < 0.001$), and 29–49 min in rural men compared with 2–17 min in urban men ($P < 0.001$). Likewise, walking at a slow pace, used as a means of transportation, was 2–3 times more often in rural *vs* urban subjects (Tables 3 and 4)

Leisure-time activity. The five most frequently reported leisure-time activities by decreasing frequency were: TV watching, discussions, reading/writing, needle work and leisure walking, in urban subjects, and discussions, singing, light farming, leisure walking and animal rearing, in rural subjects. Urban subjects spent twice the time spent by rural subjects in leisure activities ($P < 0.001$). However, light activities (< 1.5 METs) accounted for the greatest difference (Tables 3 and 4).

Total energy expenditure computed from physical activity added to non-reported time, assumed to be spent at resting metabolic rate was significantly higher in rural subjects compared to urban (Tables 3 and 4).

Physical activity, obesity, diabetes and hypertension

Obesity. There was a negative correlation between expenditure attributable to 'walk' and BMI in urban women ($r = -0.12$, $P < 0.01$), urban men ($r = -0.14$, $P < 0.01$), and rural men ($r = -0.10$, $P < 0.05$), but not in rural women. BMI was higher in the first compared to the second quartile of total expenditure in men ($\Delta = 0.75\,kg/m^2$; 95% CI 0.07–1.43). The BMI difference between the second and the third quartiles of total expenditure was significant in both women ($\Delta = 1.13\,kg/m^2$; 95% CI 0.47–1.78) and men ($\Delta = 0.76\,kg/m^2$; 95% CI 0.13–1.39). The difference between the third and the fourth quartiles of total expenditure was significant only in women ($\Delta = 0.76\,kg/m^2$; 95% CI 0.21–1.30; Figure 1).

Hypertension. Walk, leisure-time and total physical activity but not occupation negatively correlated with systolic and diastolic BP in urban men ($P < 0.01$). In rural men, systolic BP negatively correlated with walk, leisure-time activity, occupation and total activity ($P < 0.01$), and diastolic BP negatively correlated with total expenditure ($P < 0.05$). BP decreased regularly from the first to the fourth quartile of energy expenditure (Δ systolic, 3.0 mmHg in women and 9.65 mmHg in men; Δ diastolic, 1.2 mmHg in women and 4.4 mmHg in men ($P < 0.05$; Figure 1).

Diabetes. There was a significant negative correlation between physical activity (walking ($r = -0.15$), leisure ($r = -0.10$) and total expenditure ($r = -0.14$)) and fasting blood glucose in urban males (all with $P < 0.01$), but not in rural men, urban and rural women. In men, mean glycaemia decreased consistently from the first to the fourth quartile of energy expenditure ($\Delta = 0.44$ mmol/l; $P < 0.01$); a similar non-significant tendency was observed in women between the first and the third quartiles of total physical activity related energy expenditure (Figure 1).

Regression analysis. After adjustment for age, blood pressure was an independent determinant of total expenditure only in rural men: systolic ($\beta = -0.10$; adjusted $r^2 = 0.05$; $F = 13.24$; $P = 0.032$) and diastolic blood pressure ($\beta = -0.09$; adjusted $r^2 = 0.05$; $F = 12.88$; $P = 0.047$).

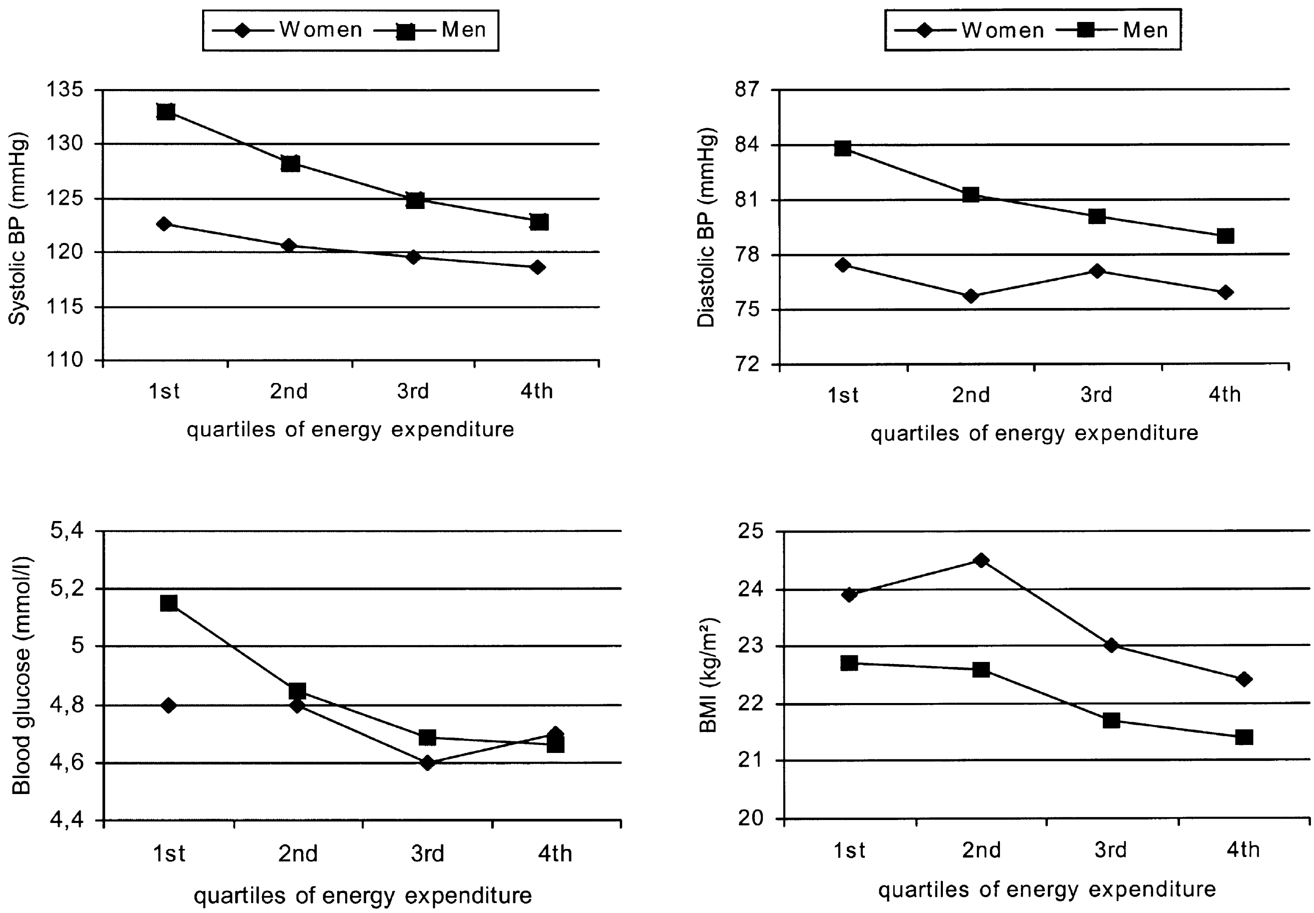

Figure 1 Blood pressure, BMI and fasting blood glucose per quartiles of energy expenditure in women and men.

Discussion

The present study carried out in urban and rural Cameroonians confirms previously reported trends in the prevalence of NCD in Sub Saharan Africa. There is a clear rural–urban positive gradient in the prevalence of obesity, diabetes and hypertension as previously reported in different populations of SSA.[13,15,27–30] Urban growth rate in Africa is estimated at 4.3% compared to 0.5% in Europe.[31] As a consequence, the number of people with type 2 diabetes mellitus in Africa, which was estimated at approximately 4.7 million in 1994, is due to triple by 2010.[32] There is also increasing evidence of the rising burden of cardiovascular diseases in Africa.[33] Differences in the prevalence of obesity, hypertension and diabetes are observed between rural and urban populations in our study. There is a five-fold gradient in obesity in both men and women and 1.5–2 times more diabetes and hypertension in urban compared to rural subjects. In a previous report, we showed that current urban residence or recent migration to an urban area were greater determinants of diabetes and cardiovascular risk than lifetime exposure to urban environment. We therefore aimed in the present study to test the hypothesis that the urban–rural difference is underlined by changes in physical activity levels, and that current levels of activity might explain the associations with recent exposure to urban lifestyle or current residence. Data from this study provide evidence for important differences in activity patterns between the two communities studied. Physical activity-related energy expenditure was higher in rural subjects compared to urban in both men and women of all age groups. Moreover, rural dwellers were mostly involved in high intensity activities whereas in the urban setting studied, both occupation and leisure-time activity were often of low to moderate intensity. Unlike urban dwellers, rural subjects exercised through long-distance walking often performed at a brisk pace as the main means of transportation. In the early 1980s, Walker *et al* studied the association between distance walked to and from school and HDL cholesterol levels in 80 South African 10–12-y-old children, and found higher levels in those who walked 10 km or more per day to get to school compared to those who lived less than 10 km from school.[34] Thus, the main difference between urban and rural dwellers may not solely be explained by the physically demanding agricultural activities as suggested by previous reports,[10,35] but also by the use of brisk-pace foot walk as the means of transportation in rural areas. Energy expenditure (in kJ) of urban dwellers equals or slightly exceeds that of their rural counterpart when body weight is taken into consideration; however, expenditure in MET is always significantly higher in the rural population. To compute energy expenditure, we used Ainsworth's compendium of metabolic cost of activities.[22] The compendium is based on measurements performed in a Western population and may reflect their activity-performance pattern. Whether these data can be used in SSA without error is therefore questionable. However, we targeted a global estimation of physical activity levels and a within-country comparison. Moreover, to reduce the potential errors, a simple scale was used based upon the type of activity performed rather than just the job title to estimate occupation-related expenditure.

Energy intake was not measured in the present study; however, a previous study in the same country reported higher intake of energy, fat and alcohol in rural men and women than in urban subjects.[36] Energy expenditure related to intense activities and frequent walking might account for the differences in BMI and prevalence of obesity observed between the urban and the rural samples studied. Indeed we observed a significant negative correlation between walking-related energy expenditure and BMI in both urban and rural subjects, and 0.75–1.13 kg/m^2 difference in BMI between the subjects who fell within the first and second quartiles of physical activity. In the BRISK Study carried out in the Cape Peninsula (South Africa), no BMI difference was observed between physically active and inactive subjects based on the assessment of occupational and leisure-time physical activity.[28] In fact, only 17% of the subjects included in that analysis had access to a car, suggesting that both groups relied mainly on walking as means of transportation.[28]

As could be anticipated, physical inactivity was associated with higher levels of blood pressure. Using the Modifiable Activity Questionnaire similar results were obtained in 799 Nigerian civil servants, of whom 500 were male.[37] These results contrast with those from the THUSA study carried out in 1821 subjects (781 males) in South Africa, where the physical activity index derived from a questionnaire based on the Baecke questionnaire was not significantly associated with blood pressure in both males and females.[38]

The negative relationship between physical activity and fasting blood glucose levels observed, mostly in urban men, is in agreement with observations in Caucasians and other Westernized populations where the deleterious effect of inactivity is clearer in men than in women.[39] However, recent investigations, whether studied cross-sectionally or following an intervention tend to show that the benefit induced by increasing levels of physical activity is almost comparable in both genders.[3,40] In our study, although the significance level was not reached in women for the benefit of physical activity against inactivity, the trend was towards the decrease in fasting blood glucose levels. The negative relationship between physical activity and diabetes was not observed in studies that used simple classification of subjects among inactive and active groups,[41,42] but has been reported in the few investigations that attempted a more precise evaluation of physical activity.[43]

When addressing the dose–response issue in the relationship between physical activity and health-related outcomes, a review of existing evidence concluded that, despite the non-standardized measurements of physical activity in population studies and the need for further investigations, physical activity tended to relate to obesity, cardiovascular disease and diabetes in an inverse dose–response fashion.[44–46] In both sexes of the urban and rural dwellers in the present study, there was a step-wise decrease in BMI, blood pressure and blood glucose levels across the increasing physical activity levels. However, the design of this study does not permit any conclusion in terms of dose–response relationship between activity and its health benefit.

In conclusion, in the subjects studied, current urban residence differed from rural by major differences in current physical activity levels, with lower occupational, walking-related and total physical activity levels. Inactivity was associated with higher BMI, blood pressure and fasting blood glucose levels. This association was more marked in men than in women. This study provides a background for a population-based intervention in attempts to prevent the rising 'epidemics' of non-communicable diseases in urban Cameroon and probably other Sub-Saharan Africa countries.

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